30 March 1966
SUPERSEDING
MIL-C-60126
26 June 1964

MILITARY SPECIFICATION

CASE, CARRYING: 8213884

1. SCOPE

1.1 This specification covers the fabrication and testing of one type plastic carrying case for prepunched program assemblies used with the Gun Direction Computer M18 system (See 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

TT-E-529	Enamel, Alkyd, Semi-Gloss
QQ-B-655	Brazing Alloy, Aluminum, and Magnesium, Filler Metal
MMM-A-132	Adhesives, Heat Resistant, Airframe Structural, Metal to Metal

Military

MIL-R-3065	Rubber, Fabricated Parts
MIL-A-5092	Adhesive, Rubber Base, General Purpose
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-W-8604	Welding of Aluminum Alloys, Process For
MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment, and Tools for Ordnance Materiel; Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

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STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-171	Finishing of Metal and Wood Surfaces
MS 15795	Washer, Flat, Metal, Round General Purpose
MS 16535	Rivet, Tubular, Oval Head

DRAWINGS

U.S. Army, Frankford Arsenal

B8582668	Decal
B8582669	Hinge
B10543502	Extrusion, Male
B10543503	Extrusion, Female
C8582671	Fastener
C10543504	Handle

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from procuring activity or as indicated by contracting officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society of Testing Materials

D256	Test for Impact Resistance of Plastics and Electrical Insulating Materials
D543	Test for Resistance of Plastics to Chemical Reagents
D638	Test for Tensile Properties of Plastics

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D648	Test for Deflection Temperature of Plastics Under Load
D785	Test for Rockwell Hardness of Plastics and Electrical Insulating Materials
D792	Test for Specific Gravity and Density of Plastics

(Copies of tests may be obtained from American Society of Testing Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 General. - Cases shall be fabricated in accordance with the requirements specified herein and shall be capable of affording physical and waterproof protection to contents. (See Figure 1 and 1A)

3.2 First article testing. - Requirements for the submission of first article samples by the contractor for testing shall be as specified in the contract.

3.3 Material. - Material used for forming case shells shall be pressed laminated sheets formed of polymers or blends of polymers, produced from acrylonitrile-butadiene-styrene. The material shall not be adulterated, extended, or in any way modified with materials other than pigments which shall not exceed 5% by weight exclusive of pigments. Materials used for processing into pressed laminated sheets shall be virgin stock. Certification of materials of each lot procured for production requirements shall be available to the Government. The property values shall meet the minimum requirements specified in Table I.

TABLE I

DETAIL REQUIREMENTS FOR RIGID ABS PLASTICS

	ASTM METHOD*	
Tensile strength, Min. psi	D638	3,500
Modulus of elasticity in tension, min. psi	D638	160,000
Rockwell hardness ("R" scale), min.	D785	80
Heat deflection, temperature under load, min. (264 psi, fiber stress)	D648	180°F
Impact strength (izod), Min. ft. lb. per inch of notch	D256 Method A	5.5
23° ± 2°C (73.4° ± 3.6°F)		2.0
-40° ± 2°C (-40° ± 3.6°F)		

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Specific Gravity		
Min.	D792	1.0
Max.	Method A	1.2
Chemical resistance 40 hour immersion in heptane 23° ± 2°C weight change, max. percent	D543	5.0

*These designations refer to the following methods of the American Society of Testing Materials:

D256	Test for Impact Resistance of Plastics and Electrical Insulating Materials
D543	Test for Resistance of Plastics to Chemical Reagents
D638	Test for Tensile Properties of Plastics
D648	Test for Deflection Temperature of Plastics Under Load
D785	Test for Rockwell Hardness of Plastics and Electrical Insulating Materials
D792	Tests for Specific Gravity and Density of Plastics

3.3.1 Color. - Material for forming case shells shall be olive drab conforming to color X24087 of FED-STD-595. Color match comparison shall be made on smooth surface. A material swatch shall be submitted for approval prior to production.

3.3.2 Finish. - Material for case shells shall have a commercial haircell grain on the exterior surface and a haircell or smooth finish on the interior surface.

3.3.3 Fungus resistance. - The material used for forming case shells shall show no evidence of damage, deterioration or profuse growth following an incubation period of 28 days after inoculation of surfaces with a mixed spore suspension.

3.4 Case forming process. - Case shells shown on Figures 2 and 3 shall be thermo-formed by any proven method normally associated with the plastic industry. Finished formed parts shall be entirely free of blisters, and blemishes, with no vacuum vent marks to be visible on outside surfaces of the formed pieces.

3.5 Fabrication.

3.5.1 Sealing extrusions. - The upper sealing extrusion (male) Drawing B10543502 and the lower sealing extrusion (female) Drawing B10543503 shall be formed in one continuous operation by proven forming techniques which shall assure uniform fit over case shells and shall fit freely in each other when the case is closed. The formed extrusions shall be free of chatter, tool marks and distortions. The extrusions shall be joined by butt welding or brazing which shall be ground smooth and exhibit soundness

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of good practices in accordance with MIL-W-8604 and QQ-B-655.

3.5.1.1 Finish. - Prior to assembly to case shells, the extrusions shall be treated by finish 7.3, followed by paint finish 21.3 of MIL-STD-171, color X24087 of FED-STD-595. (See Figure 4)

3.5.1.2 Bonding of extrusions. - The extrusions shall be attached to the case shells with an air curable, two part, epoxy adhesive conforming to MMM-A-132, Type I, Class 3, applied in a manner to provide structural integrity and a watertight joint. Accellerated cure by heating shall not exceed 145 ± 5°F, for a period of 5 hours. (See Figure 1A)

3.5.2 Closure fasteners. - Two (2) fasteners, part no. 8582671-1 shall be assembled to the case as shown on Figure 1. Rivet heads shall be touched up with paint conforming to TT-E-529, color X24087 of FED-STD-595.

3.5.3 Hinge. - The body and the cover shall be attached by two (2) hinges part no. 8582669-1, assembled as shown on Figure 1. Rivet heads shall be touched up with paint conforming to TT-E-529, color X24087 of FED-STD-595.

3.5.4 Decal. - A pressure sensitive decal conforming to Drawing B8582668 shall be installed as shown on Figure 1A. Prior to assembly of decal, the finish of the case shall be slightly buffed to prepare surface for assembly.

3.5.5 Handle. - One (1) handle assembly conforming to Drawing B10543504 shall be assembled to the case as shown on Figure 1.

3.5.6 Sealing gasket. - Gasket shall be fabricated from rubber, .156 O.D. by .093 I.D., durometer 40-60, conforming to MIL-R-3065, Class SC, Grade 412ANFFK. The gasket may be formed by molding, vulcanizing or by butt bonding. Butt bonding shall be accomplished with adhesive conforming to MIL-A-5092, Type II. Prior to inserting the gasket in the female extrusion, adhesive conforming to MIL-A-5092, Type II shall be uniformly applied within the well. There shall be no evidence of excess adhesive in the gasket well. (See Figure 1A)

3.6 Environmental.

3.6.1 Impact resistance. - The case assembly shall show no evidence of cracking, chipping or delamination when dropped from a height of 30 inches.

3.6.2 Waterproofness. - The case assembly shall show no evidence of leakage when submerged under 1 inch of water for one minute.

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3.7 Performance. - The fasteners, hinges used to secure the cover and the handle shall be correctly positioned and shall function without interference. The hinge shall permit opening and closing without binding.

3.8 Marking. - Marking shall be applied as shown on Figure 1 by silk screen or stencil, system 21.1 of MIL-STD-171; color white 17875 of FED-STD-595.

3.9 Workmanship. - Cases fabricated to the requirements of this specification shall be in accordance with proven commercial forming techniques and shall be free of defects that might affect strength, serviceability and protection of contents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility of inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article testing. - When the contract specifies testing of preproduction or initial production samples, the following paragraphs shall apply as applicable.

4.2.1 Preproduction test. - When specified in the contract, two preproduction samples shall be submitted to the procuring agency for approval of design and construction features. Preproduction samples shall meet all inspection provisions of 4.3 and 4.6 as well as any additional tests that may be specified in the contract.

4.2.2 Initial production test. - Initial production test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, samples shall be the first two production units and shall have been manufactured using the same materials, processes and procedures as will be used in regular production. Initial production testing shall be in accordance with 4.3, 4.4 and 4.6 and shall be witnessed by the Government representative for all examinations and tests. The Government reserves the right to conduct any or all examinations or tests as may be deemed necessary.

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4.2.3 First article disapproval. - Failure of any one of a first article sample submitted shall be cause for disapproval by the Government and subsequent action as required by the contract clauses applicable to first article testing.

4.3 Inspection provisions.

4.3.1 Submission of product. - Unless otherwise specified by the Contracting Officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests. - Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926, and the Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents (IEL) when provided with the technical data package (TDP). Examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105, and the sampling plans specified in Table II. Examinations and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot.

TABLE II - CLASSIFICATION OF DEFECTS

CLASS CHARACTERISTIC		REQUIREMENT	TEST PROCEDURES
CRITICAL: NONE			
MAJOR: AQL 1.0% defective			
101.	Improper forming	3.4	Visual
102.	Sealing extrusion improperly formed, welded or brazed	3.5.1	Visual
103.	Improper paint finish	3.5.1.1	Visual
104.	Improper bonding of extrusion, improper sealant	3.5.1.2	Visual
105.	Improper fasteners or assembly to case	3.5.2	Visual
106.	Improper hinge or assembly to case	3.5.3	Visual
107.	Improper decal or poor adhesion	3.5.4	Visual
108.	Improper handle or improper assembly	3.5.5	Visual
109.	Improper gasket, improperly secured	3.5.6	Visual
110.	Marking incorrect or missing	3.8	Visual

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MINOR: AQL 4.0% defective

201.	Color match (smooth surface)	3.3.1	Visual
202.	Improper finish	3.3.2	Visual
203.	Excess adhesive on gasket	3.5.6	Visual

4.3.3 Acceptance and rejection. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3.4 Certification. - Certification shall include test data and results for all characteristics specified below. Certification will be required prior to performing final acceptance inspection and shall suffice for the remainder of the production quantities, provided the manufacturing processes and techniques used to produce the items for which certification was issued have not been changed or revised. Any and all changes will require a new certification from the contractor.

No.	CHARACTERISTIC	TEST DATA TO COMPLY WITH
301.	Material	3.3

4.4 Special sampling. -

4.4.1 General. - One case shall be selected at random as a control sample from each 100 produced, or from each month's production, whichever comes first. The control sample shall have successfully met all other acceptance test specified herein, and shall then meet the following requirements and tests:

CONTROL TEST		REQUIREMENT	TEST PROCEDURE
401.	Fabrication	3.5 - 3.8	SME, Visual Figures 1, 1A, 2, 3, and 4
402.	Workmanship	3.9	Visual, tactile

4.4.2 Environmental and functional. - Two cases shall be selected at random as control samples from each 100 produced, or from each month's production, whichever occurs first. The control samples shall have successfully met all other acceptance tests specified herein, and shall then meet the following requirements and tests:

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CONTROL TEST	REQUIREMENT	TEST PROCEDURE
403. Impact resistance	3.6.1	4.6.2
404. Waterproofness	3.6.2	4.6.4
405. Performance	3.7	4.7.1

4.4.3 Failure of sample. - Should any one item of a control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.5 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment. - Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. - Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents (IEL), when provided with the Technical Data Package (TDP).

4.5.2.2 Contractor design. - The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures.

4.6.1 Preparation for tests. - The case shall be fitted with a dummy load having a mass 10 by 9 by 8, having a weight of 10 pounds. The dummy load shall be adequately immobilized when the cover is closed and secured.

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4.6.2 Impact resistance. - The sample fitted with dummy load shall be dropped from a height of 30 inches on to a concrete or steel surface, once on the top and once on the bottom. Following the test, the case shall be inspected for evidence of failure in accordance with 3.6.1.

4.6.3 Fungus resistance. - Tests shall conform to fungus resistance test, Procedure I of MIL-E-5272. Sample material shall be minimum of one square foot. At the completion of the test period, any signs of damage, deterioration or profuse growth shall consititute cause for rejection. The fungus test shall not be required for suppliers having qualified by certification that their material has been tested by above procedure and is fungus resistant.

4.6.4 Waterproofness. - The case, conditioned at ambient temperatures shall be immersed in water so that the top surface is one inch below the surface of the water for a period of one minute. After removal, the case shall be opened and inspected for water leakage.

4.6.4.1 Interpretation of results. - Evidence of water leakage shall be cause for rejection.

4.7 Performance tests.

4.7.1 External components. - This shall be a visual and manual check, and shall be in accordance with the requirements specified. The fasteners used to secure the cover of the case, shall function without interference and shall show no evidence of looseness. The handle shall support the case with inclosed dummy load when carried. The cover shall open and close without restriction or binding. The gasket shall be examined for alignment and fit in sealing extrusion well.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. - Unless specified, the cases shall be prepared for shipment using normal commercial methods, or materials to assure acceptance by common or other carrier for safe transportation at the lowest rate to point of destination.

5.2 Marking. - Marking for shipment shall be in accordance with MIL-STD-129.

6 NOTES

6.1 Intended use. - The case is intended to accommodate a maximum of eight cartridge assemblies, with program tapes and manuals to be used in conjunction with the Signed Data Reproducer RP-143/GSQ-64 or the Test, Computer Logic Unit AN/GSM-70.

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6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Applicable stock number.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1220-A245Y

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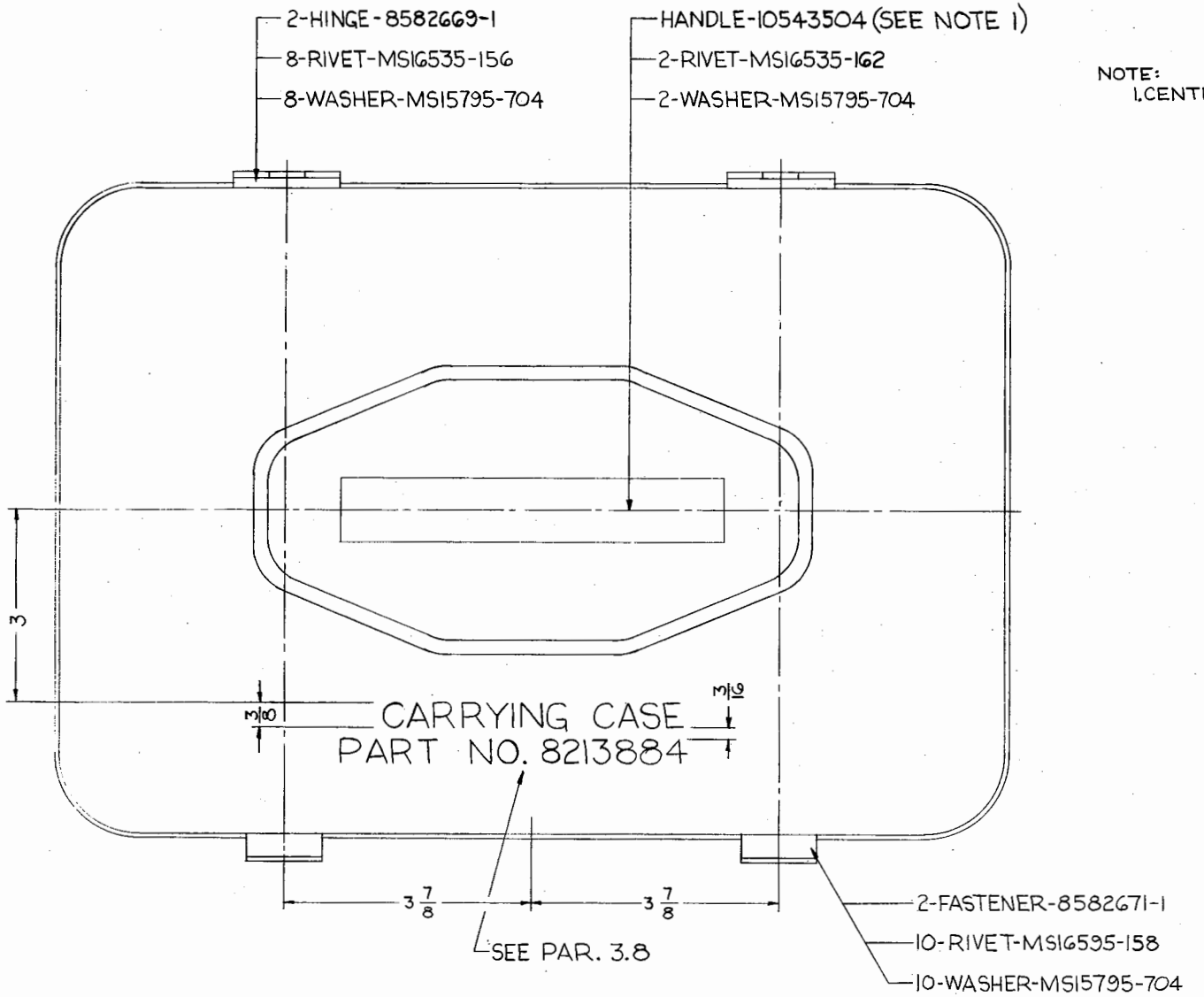


FIGURE 1-CASE ASSEMBLY

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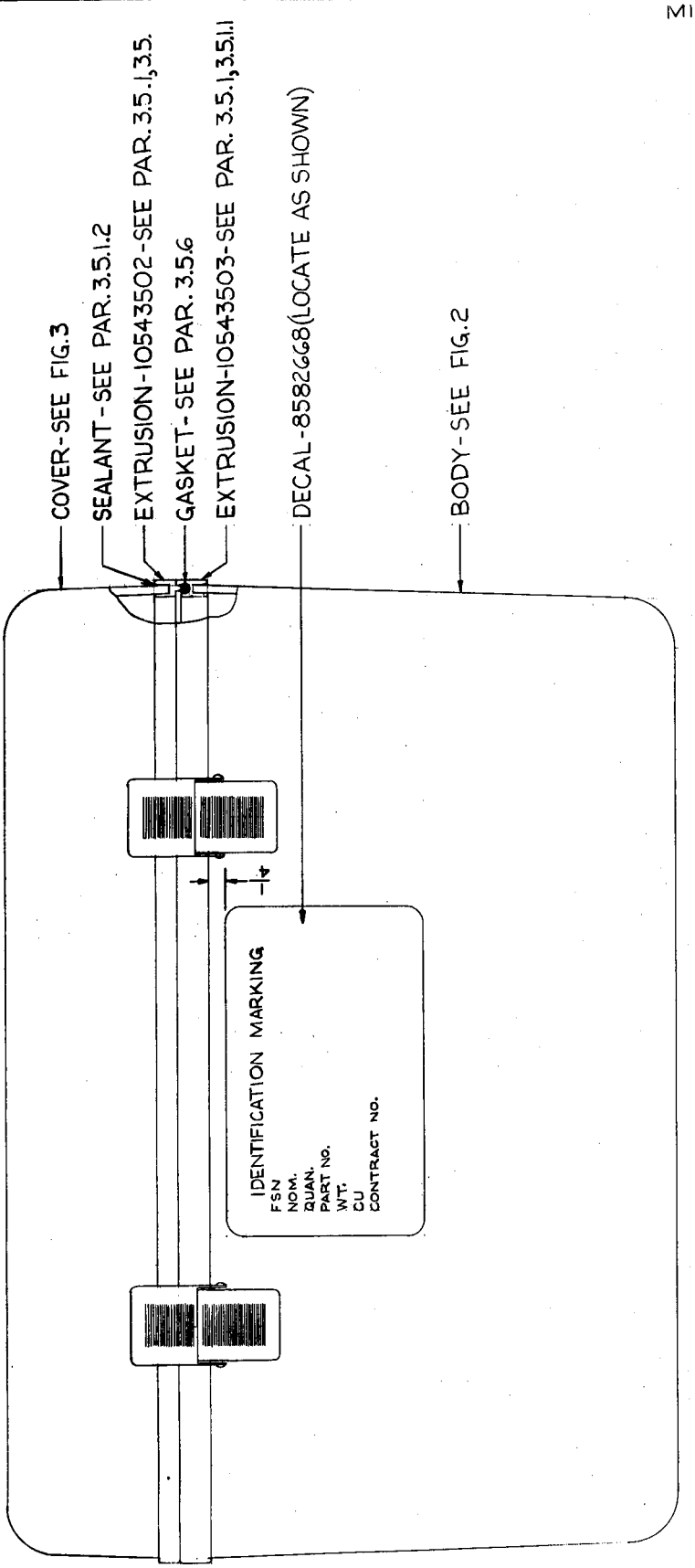


FIGURE IA-CASE ASSEMBLY

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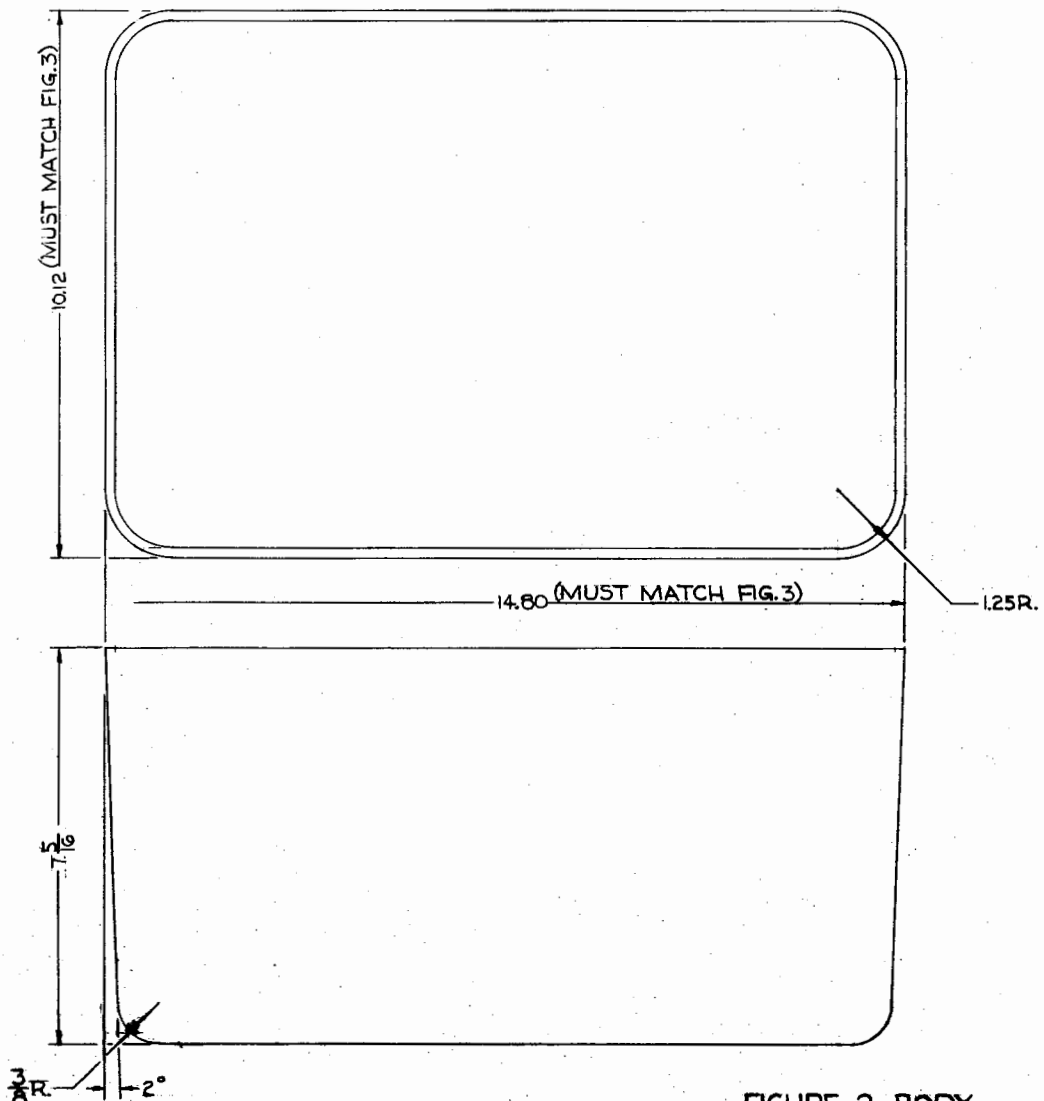
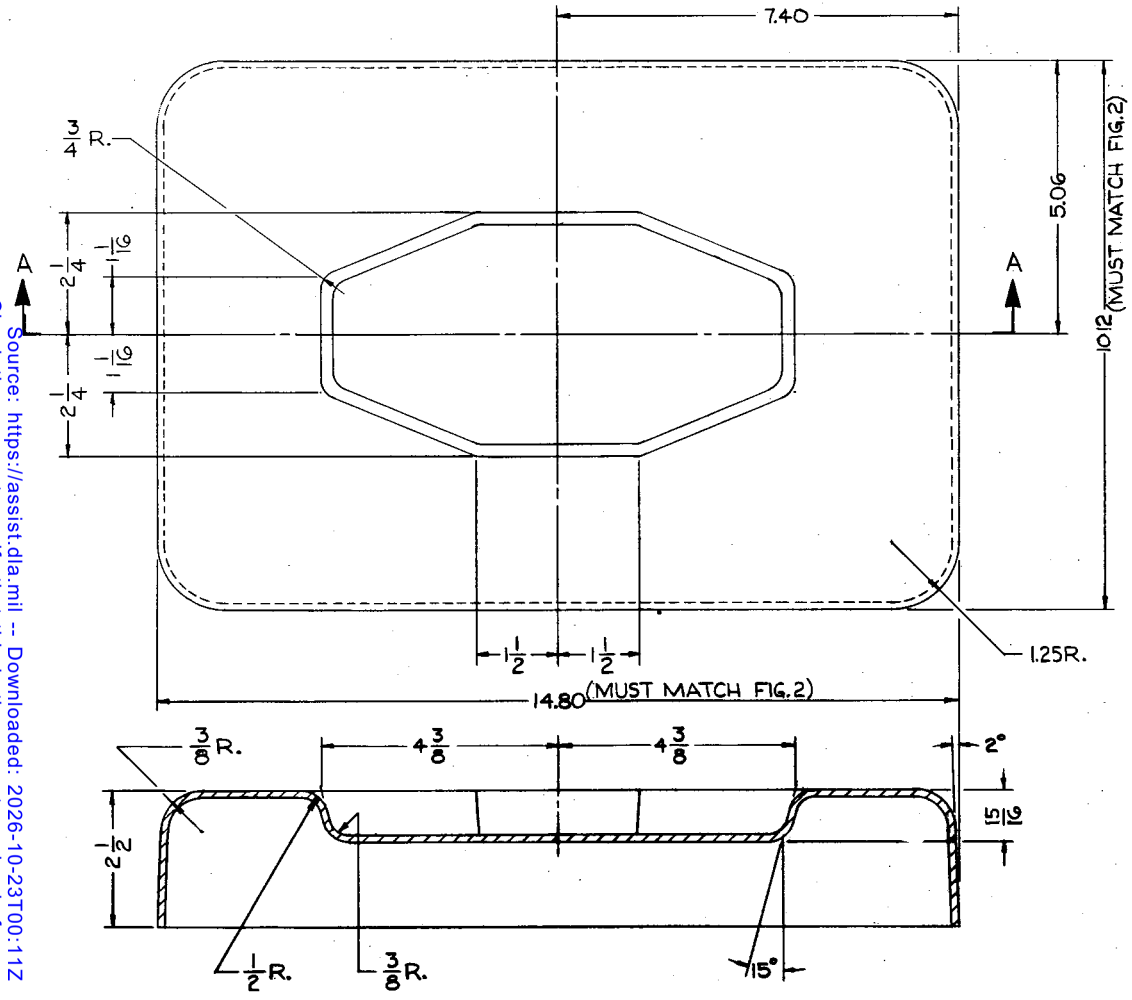


FIGURE 2 - BODY

NOTES:

1. MATERIAL-ABS PLASTIC, PER PAR. 3.3, COLOR OLIVE DRAB, X24087 OF FED-STD-595.
   APPROVED SOURCES:
   U.S. RUBBER CO., CHICAGO, ILLINOIS
   PART NO. R40- -CPA1492
   OR
   BOLTA PRODUCTS DIV. OF GEN'L TIRE & RUBBER CO., LAWRENCE, MASS.
   BOLTARON 6000 SERIES, X4451
   STARTING THICKNESS 3/16 INCH
2. TOLERANCES- ANGLES ±0° 30'
   FRACTIONS ±1/32
   DECIMALS ±.03

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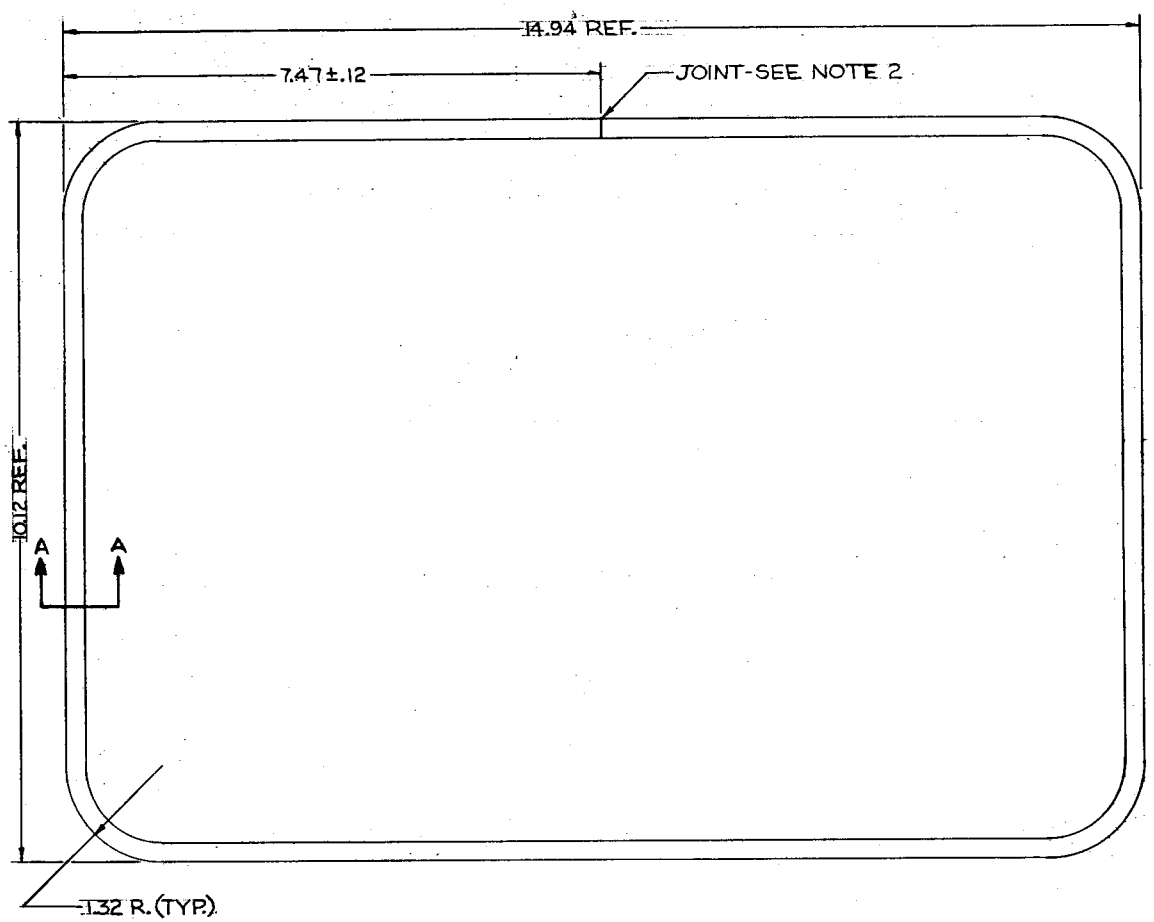
SECTION A-A

NOTES:

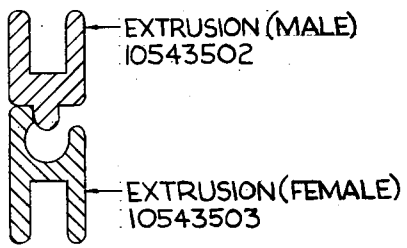
1. MATERIAL-ABS PLASTIC, PER PAR. 3.3, COLOR-OLIVE DRAB, X24087 OF FED-STD-595.
   APPROVED SOURCES
   U.S. RUBBER CO., CHICAGO, ILLINOIS
   PART NO. R40- -CPA1492
   OR
   BOLTA PRODUCTS DIV. OF GEN'L TIRE & RUBBER CO., LAWRENCE, MASS.
   BOLTARON 6000 SERIES, X4451
   STARTING THICKNESS 1/8 INCH
2. TOLERANCES - ANGLES ±0° 30'
   FRACTIONS ±1/32
   DECIMALS ±.03

FIGURE 3. COVER

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NOTES:
1. COMPLETED EDGE FRAMES SHALL FIT FREELY INTO EACH OTHER AS SHOWN IN SECTION "A-A" & MUST MATCH SHELLS SHOWN IN FIGURES 2 & 3.
2. EXTRUSIONS SHALL BE FORMED, WELDED OR BRAZED, & FINISHED IN ACCORDANCE WITH PAR. 3.5.1, 3.5.1.1



SECTION A-A

FIGURE 4-EXTRUSION ASSEMBLIES

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